

Project Title

E-Commerce Order Management Platform

Problem Statement

Design and develop an **E-Commerce Order Management Platform** to manage products, customers, orders, inventory, and payments with secure payment gateway integration, scalable client-server architecture, concurrent order processing, transaction management, and database-driven operations.

Common Responsibilities (All Teams)

Before beginning development, all teams must jointly finalize:

- Software Requirement Specification (SRS)
 - ER Diagram
 - Database Schema
 - Table Names and Primary Keys
 - Coding Standards
 - Folder Structure
 - Naming Conventions
 - API/Function Interfaces
 - Git Repository Structure (if applicable)
 - UI Design Standards
 - System Architecture
 - Testing Strategy
-

TEAM 1

Worklet: Customer & Product Management

Objective

Develop the core customer and product management module responsible for maintaining users, customer accounts, products, categories, brands, and product information.

Major Modules

- User Registration
 - Login Authentication
 - Role-Based Access Control
 - Customer Management
 - Product Management
 - Category Management
 - Brand Management
 - Product Search
 - Product Reviews & Ratings
 - Product Image Management
-

Database Tables

- Users
 - Roles
 - Customers
 - Products
 - Categories
 - Brands
 - ProductReviews
-

Subject-wise Implementation

Subject	Expected Implementation
Basic Programming	Modular program design, product validation, file handling
Object-Oriented Programming	User, Customer, Product, Category and Brand classes
Data Structures	Arrays/Linked Lists for product storage and searching
DBMS	CRUD operations, SQL queries, constraints

Computer Networks	Client-server login request handling
Operating Systems	Product image and file management
Software Engineering	SRS, UML, Use Case Diagram, Class Diagram
Web Technology	Login Page, Customer Dashboard, Product Catalog

Deliverables

- Login Module
 - Customer Management Module
 - Product Management Module
 - Category Module
 - Brand Module
 - Database Scripts
 - UML Diagrams
 - Unit Testing Report
-

TEAM 2

Worklet: Order & Inventory Management

Objective

Develop the order processing and inventory management module responsible for shopping cart operations, order processing, inventory control, and concurrent order handling.

Major Modules

- Shopping Cart Management
- Wishlist Management
- Order Placement
- Order Processing
- Order Tracking
- Order Cancellation
- Inventory Management

- Stock Management
 - Invoice Generation
 - Concurrent Order Processing
-

Database Tables

- Cart
 - Wishlist
 - Orders
 - OrderItems
 - Inventory
 - Warehouses
 - OrderStatus
-

Subject-wise Implementation

Subject	Expected Implementation
Basic Programming	Order processing and inventory update logic
Object-Oriented Programming	Cart, Order, Inventory and Invoice classes
Data Structures	Queue/Linked List for order processing
DBMS	Order and Inventory CRUD operations
Computer Networks	Multi-user order request handling
Operating Systems	Thread synchronization for concurrent order processing
Software Engineering	Activity Diagram, Sequence Diagram
Web Technology	Shopping Cart, Checkout and Order Pages

Deliverables

- Shopping Cart Module
- Order Management Module
- Inventory Management Module
- Invoice Generation Module
- Concurrency Demonstration

- Database Scripts
 - UML Diagrams
 - Unit Testing Report
-

TEAM 3

Worklet: Payment, Reports, Analytics & System Administration

Objective

Develop payment processing, reporting, analytics, and administrative features of the platform and integrate all project modules.

Major Modules

- Payment Gateway Integration
 - Payment Processing
 - Transaction Management
 - Refund Processing
 - Sales Reports
 - Customer Reports
 - Dashboard & Analytics
 - Notifications
 - Audit Logs
 - Export Reports
 - Backup & Restore
 - System Monitoring
-

Database Tables

- Payments
- Transactions
- Reports
- Notifications
- AuditLogs

- BackupHistory

Subject-wise Implementation

Subject	Expected Implementation
Basic Programming	Payment validation and report generation
Object-Oriented Programming	Payment, Transaction, Report and Notification classes
Data Structures	Queue/Stack for payment requests and notifications
DBMS	Transaction management, aggregate queries, report generation
Computer Networks	Secure payment gateway communication
Operating Systems	Backup scheduling and log management
Software Engineering	Integration Testing, Documentation
Web Technology	Dashboard, Reports, Payment Pages, Export Pages

Deliverables

- Payment Module
 - Dashboard
 - Reports Module
 - Notification Module
 - Audit Log Module
 - Backup & Restore Module
 - Integration Report
 - Testing Report
-

Key Performance Indicator (KPI)

Sr. No.	Key Performance Indicator (KPI)	Target
---------	---------------------------------	--------

1	Functional Completeness	100% implementation of all modules as per the problem statement
2	Automation Coverage	100% automation of customer management, product management, order processing, inventory management, payment processing, and report generation
3	User Authentication & Role-Based Access	Secure login with 100% role-based authorization for Customers, Administrators, and Vendors
4	Database Integrity	100% data consistency with no duplicate or orphan records
5	System Response Time	Average response time ≤ 2 seconds for standard operations
6	Concurrent Order Handling	Support at least 20 concurrent order requests without race conditions, crashes, or data inconsistency
7	Inventory Accuracy	100% accurate stock updates after every order, cancellation, and return transaction
8	Payment Transaction Accuracy	100% successful payment recording and secure transaction validation
9	Code Quality & Documentation	Modular, well-documented code with complete SRS, UML diagrams, ER diagram, database schema, API documentation, and testing reports
10	System Integration & Reliability	Successful integration of all three worklets with no critical runtime errors during final demonstration

Expected Final Outcome

At the end of the semester, the integrated platform should support:

- Secure Login with Role-Based Access
- Customer Registration & Profile Management
- Product & Category Management
- Brand Management
- Product Search, Filtering & Reviews
- Shopping Cart & Wishlist Management
- Order Placement, Tracking & Cancellation
- Concurrent Order Processing
- Inventory Management with Real-Time Stock Updates
- Secure Payment Gateway Integration

- Invoice Generation & Payment History
- Sales Reports & Analytics Dashboard
- Notifications & Order Status Updates
- Audit Logs
- Backup & Restore
- Database-Driven Operations
- Complete Documentation and Testing